DECIDABILITY AND REDUCIBILITY

At this point, we need to look back to page 6 of week #2 on context free languages in order to learn about LR(1) reduction and decidability. We will repeat the exercise we did then, but with a different grammar as well as a "short cut" to the all-important Shift-Reduce matrix. This will set us up for Special Problem Set #4.

Consider the following grammar:

0.1 X ---> A
1.1 A ---> aB
1.2 A ---> CD
2.1 B ---> b
3.1 C ---> aD
3.2 C ---> BA
4.1 D ---> d

The "short cut" referred to above will require the following table headings:

STATE	ENTRY	ITEMS	NEXT	SR ENTRY

We will start with state 0 which will look like this:

0	1	X ---> ^A	1	0,A = 1

Since the marker is in front of the non-terminal A we will add all the productions for A with the marker at the far left as follows

2	A ---> ^aB	2	0,a = 2
3	A ---> ^CD	3	0,C = 3

Now, in the last entry, the marker is on front of the non-terminal C so we must add all of the productions for C with the marker at the far left as follows:

4	C ---> ^aD	2	0,a = 2

You should note that the reason NEXT is 2 and not 4 is because we already indicated in entry 2 that we would go to state 2 when we were marking "a"

5	C ---> ^BA	4	0,B = 4

And now we must add the productions, there is only one, on B:

6	B ---> ^b	5	0,b = 5

We have now completed the entries for state 0. We go to state 1

1	1	X ---> A^	Complete	1,FB(X) = 1,# =accept

State 2 will have four entries. For those entries in state 0 where NEXT = 2 we add them to STATE 2 with the marker moved one symbol to the right. This will cause the marker to be in front of both B and D. So we have:

2	1	A ---> a^B	6	2,B = 6

Now we must add the productions on B with the marker at the beginning:

2	B ---> ^b	5	2,b = 5

Why the 5? Look at state 0 entry number 6. Continuing:

3	C ---> a^D	7	2,D = 7
4	D ---> ^d	8	2,d = 8

And so forth! Can you finish this derivation of the SR entries?

Attached you will find the BW matrix, the FB matrix, and the SR matrix for this grammar. You will also find what at first will look like an old-fashioned flow chart. Actually it shows all the transitions between the states and will be used in Special Problem Set #4.

CS 475 SPECIAL PROBLEM SET #4

For the input b a b d # construct a trace like the one that begins in the middle of page 8 of the posting in unit #2.

Also, by using the numbers in the boxes in the "flow chart" … track the process of the input string b a b d # as it "moves" through the chart. Here is a start that will hopefully give you an idea of what the problem is asking you to do. The underscore is showing you where the file pointer is pointing. The "boxed" numbers are at the left without the box notation. We have the following "start":

0: b a b d #

1: b a b d #

0: B a b d #

4: B a b d #

2: B a b d #

There is about a dozen or so more to go. Note: The ovals in the chart represent completions, and as you know that causes "you" to pop the stack a number of times. Popping corresponds to backtracking through the chart by the same number of steps that Pushed you to wherever you were. Following the arrowheads is equivalent to Pushing onto the stack and following them backwards is equivalent to Popping the stack. Hopefully you will end up at the oval labeled by box number one! Also, every time a terminal or a string of terminals is replaced by a non-terminal we have a "simple" REDUCTION.

All of what we have done comes under the "simple" concept of DECIDABILITY. I guess it all comes down to what is meant by SIMPLE!

GIVEN:

X---->A
A----->aB
-----><u>CD</u>
B----->b
C----->aD
-----><u>BA</u>
D----->d

(1) MARKER

(2) EMBEDDED

(3) END

COMPLETE THE BW MATRIX:

	X	A	B	C	D	a	b	d
X		/	*	*		*	*	
A			*	/		/	*	
B							/	
C			/			/	*	
D								*

COMPLETE THE FB MATRIX

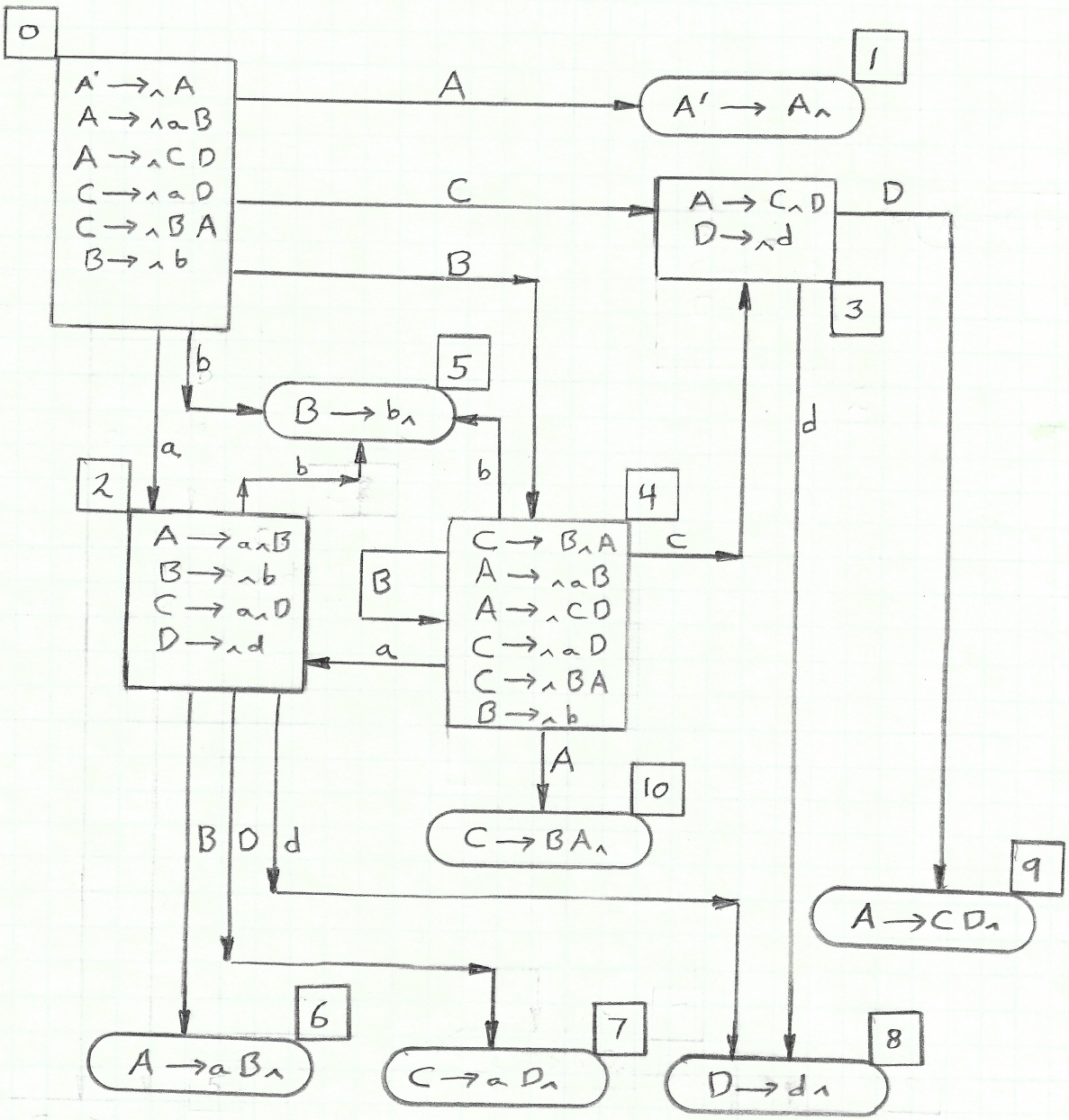
	X	A	B	C	D	a	b	d	#
X									1
A								3	3
B						2	2	3	3
C								2	
D								3	3

A SIMPLE GRAMMAR

1.1	A->aB		
1.2	A->CD		
2.1	B->b		
3.1	C->aD		
3.2	C->BA		
4.1	D->d.		

THE SHIFT-REDUCE MATRIX FOR THE SIMPLE GRAMMAR

	A	B	C	D	a	b	d	#
0	1	4	3		2	5		
1								accept
2		6		7		5	8	
3				9			8	
4	10	4	3		2	5		
5					2.1	2.1	2.1	2.1
6							1.1	1.1
7							3.1	
8							4.1	4.1
9							1.2	1.2
10							3.2	


0
A' → ^A
A → ^aB
A → ^CD
C → ^aD
C → ^BA
B → ^b
A
1
A' → A^
C
A → C^D
D → ^d
D
3
B
b
5
B → b^
a
b
b
2
4
A → a^B
B → ^b
C → a^D
D → ^d
C → B^A
A → ^aB
A → ^CD
C → ^aD
C → ^BA
B → ^b
B
a
c
d
A
10
C → BA^
B
D
d
9
A → CD^
6
A → aB^
7
C → aD^
8
D → d^